

Aniruddhan Narasimhan

aniruddhan26@gmail.com | (202)-352-2975 | LinkedIn | GitHub | Portfolio | DataFolio

Education

University of Maryland – College Park

M.S. in Applied Machine Learning

Aug 2025 – May 2027

GPA: 3.9/4.0

Vellore Institute of Technology (VIT), Chennai

B.Tech. in Computer Science and Engineering (AI & Robotics)

2021 – May 2025

GPA: 3.6/4.0

Experience

Machine Learning Intern – KCF Technologies

Jun 2026 – Aug 2026

(Hybrid) State College, PA, USA

- Developed and evaluated **reinforcement learning-based fault classification models using PPO and DQN** across **30M+ industrial vibration sensor records**, multiple asset types, monitoring points, and X/Y vibration axes.
- Built scalable data preprocessing, experimentation, and inference workflows using **Python, PySpark, Databricks, and MLflow**, supporting missing-data analysis, feature preparation, asset-level validation, experiment tracking, and model inference.
- Identified and mitigated **temporal data leakage and class imbalance** across data splitting, imputation, and feature scaling, improving **unhealthy-class recall by 20–30%** while establishing a robust fault-classification pipeline evaluated using precision, recall, F1-score, and confusion matrices.

Trainee & Project Contributor – FincruX Technologies LLP

May 2024 – Aug 2024

Chennai, India

- Built predictive modeling pipelines for digital advertising data using **Python and statistical modeling**, improving predictive accuracy by **7%** through regularization, preprocessing, and data validation.
- Analyzed **time-series trends and advertising performance data** to identify revenue-impacting patterns supporting data-driven campaign and targeting decisions.

ASO Intern – MUFG Global Services Pvt. Ltd.

Nov 2023 – Dec 2023

Bengaluru, India

- Developed and deployed an organization-wide leave management application using **Microsoft Viva, SharePoint Framework, Microsoft Teams, and Office 365**.
- Automated approval workflows using **Power Automate**, reducing manual follow-ups by **60%** and improving request-handling efficiency by **40%**.

Research Experience

Research Author & Contributor – Vellore Institute of Technology, Chennai

Dec 2024 – Jun 2025

Dr. K. Valarmathi

- Contributed to **four machine learning research works** spanning medical imaging, quantum-driven feature extraction, ensemble learning, time-series forecasting, pancreatic cancer detection, and obesity-risk classification
- Co-authored **Enhanced Brain Tumour Prediction Using Quantum: A Hybrid Deep Learning Approach**, published in *Scientific Reports* (2026), and **Scalable Enhancement of Cloud-Based DDoS Detection with Adaptive ML Pipelines**, available on *TechRxiv*.

Projects

Brain Tumor Detection using ResNet-50

- Implemented and fine-tuned a **ResNet-50** image classification pipeline, improving accuracy by **12% over VGG-19** and achieving **99% AUC** for tumor versus non-tumor MRI classification.
- Improved model robustness through **data augmentation, class balancing, preprocessing, and threshold optimization** using TensorFlow, Keras, Scikit-learn, and OpenCV.

AI-Powered Nutrition Analyzer

- Developed an AI-driven nutrition application using **Gemini Vision Pro** for automated meal recognition, macro-nutrient estimation, personalized recommendations, and dietary tracking.
- Built and deployed the application using **Python, Gemini API, Git, and Streamlit Community Cloud**.

Technical Skills

Languages: Python, SQL, R, C++

Machine Learning: Scikit-learn, PyTorch, TensorFlow, Keras, Pandas, NumPy, Time Series, Deep Learning, Statistical Modeling

GenAI: LangChain, LlamaIndex, RAG

Data & MLOps: PySpark, Databricks, Git, FastAPI, DVC, MLflow, Docker, Git, GitHub Actions, CI/CD

Cloud & Deployment: AWS, Azure, Vertex AI, Flask, Streamlit, Gradio

Databases: PostgreSQL, Microsoft SQL Server